

Attention Is All You Need

Ashish Vaswani, Noam Shazeer, Niki Parmar

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks. We propose the Transformer, a new simple network architecture based solely on attention mechanisms. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable.

Introduction

Recurrent neural networks have been firmly established as state of the art approaches in sequence modeling. This inherently sequential nature precludes parallelization within training examples.

Results

On the WMT 2014 English-to-German translation task, the Transformer achieves a BLEU score of 28.4, improving over the best previously reported models by over 2 BLEU.

Conclusion

We presented the Transformer, the first sequence transduction model based entirely on attention. It can be trained significantly faster than architectures based on recurrent or convolutional layers.